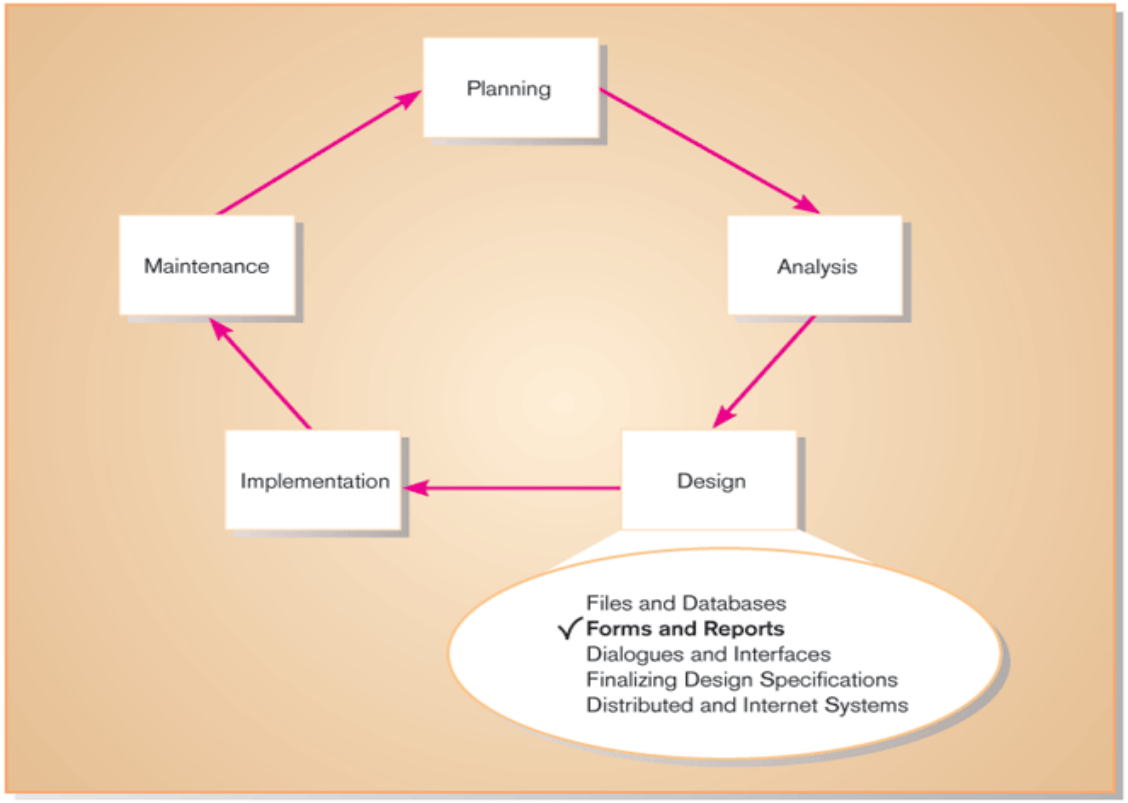
Unit-4
Design

DESIGNING FORMS AND REPORTS

Designing Forms and Reports

Figure 11-1 Systems development life cycle with logical design phase highlighted



Designing Forms and Reports (Cont.)

Form: a business document that contains some predefined data and may include some areas where additional data are to be filled in.

- An instance of a form is typically based on one database record.

Designing Forms and Reports (Cont.)

Report: A business document that contains only predefined data;

- It is a passive document used solely for reading or viewing data.

A report typically contains data from many unrelated records or transactions.

Designing Forms and Reports (Cont.)

Common Types of Reports:

- *Scheduled:* produced at predefined time intervals for routine information needs.
- *Key-indicator*: provide summary of critical information on regular basis.

Designing Forms and Reports (Cont.)

- *Exception*: highlights data outside of normal operating ranges.
- *Drill-down*: provide details behind summary of key-indicator or exception reports.
- *Ad-hoc*: respond to unplanned requests for non-routine information needs.

The Process of Designing Forms and Reports

User-focused activity.

Follows a prototyping approach.

First steps are to gain an understanding of the intended user and task objectives by collecting initial requirements during requirements determination.

The Process of Designing Forms and Reports

Requirements determination:

- Who will use the form or report?
- What is the purpose of the form or report?
- When is the report needed or used?
- Where does the form or report need to be delivered and used?
- How many people need to use or view the form or report?

The Process of Designing Forms and Reports (Cont.)

Prototyping

- Initial prototype is designed from requirements.
- Users review prototype design and either accept the design or request changes.
- If changes are requested, the construction-evaluation-refinement cycle is repeated until the design is accepted.

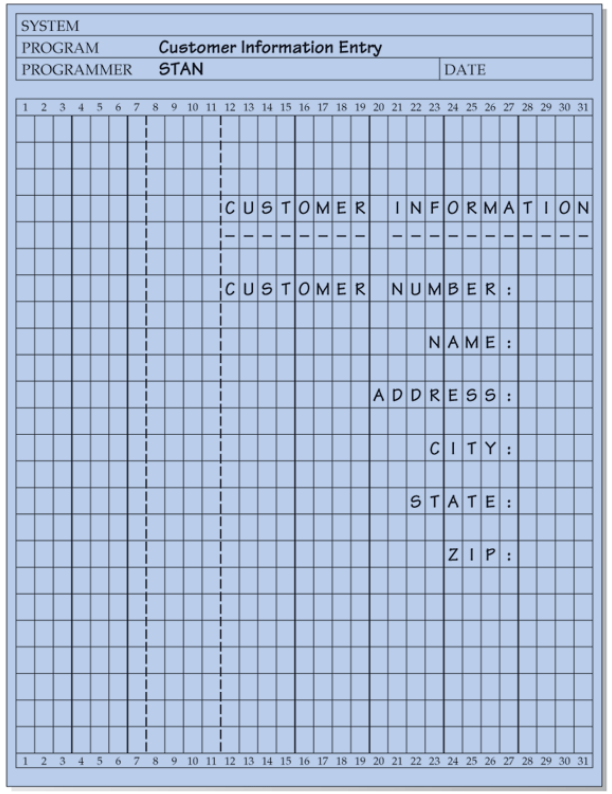
The Process of Designing Forms and Reports (Cont.)

A coding sheet is an “old” tool for designing forms and reports, usually associated with text-based forms and reports for mainframe applications.

Visual Basic and other development tools provide computer aided GUI form and report generation.

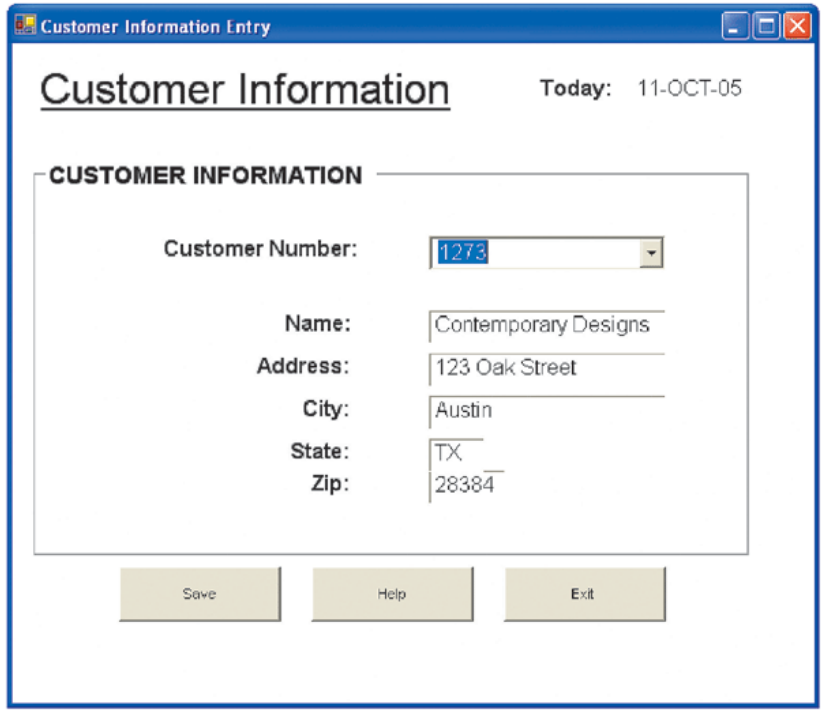
The Process of Designing Forms and Reports (Cont.)

Figure 11-2 The layout of a data input form using a coding sheet



The Process of Designing Forms and Reports (Cont.)

Figure 11-3
A data input screen designed in Microsoft's Visual Basic .NET



Deliverables and Outcomes

Design specifications are the major deliverables and inputs to the system implementation phase.

Design specifications have three sections:

- *Narrative overview*: characterizes users, tasks, system, and environmental factors.

Deliverables and Outcomes (Cont.)

- *Sample design*: image of the form (from coding sheet or form building development tool).
- Testing and usability assessment: measuring test/usability results (consistency, sufficiency, accuracy, etc.).

Formatting Forms and Reports

Meaningful titles: clear, specific, version information, current date.

Meaningful information: include only necessary information, with no need to modify.

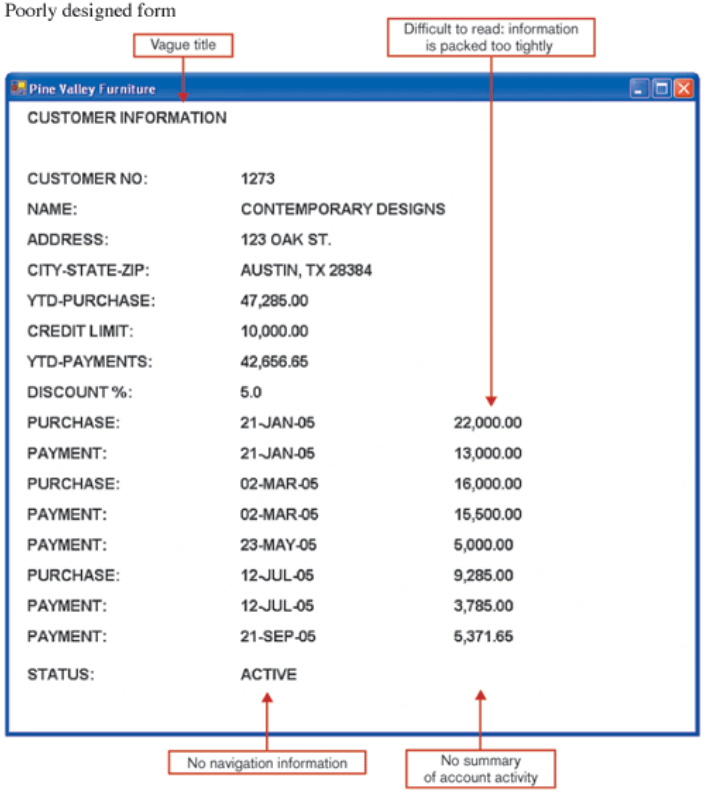
Formatting Forms and Reports (Cont.)

Balanced layout: adequate spacing, margins and clear labels.

Easy navigation system: show how to move forward and backward, and where you are currently.

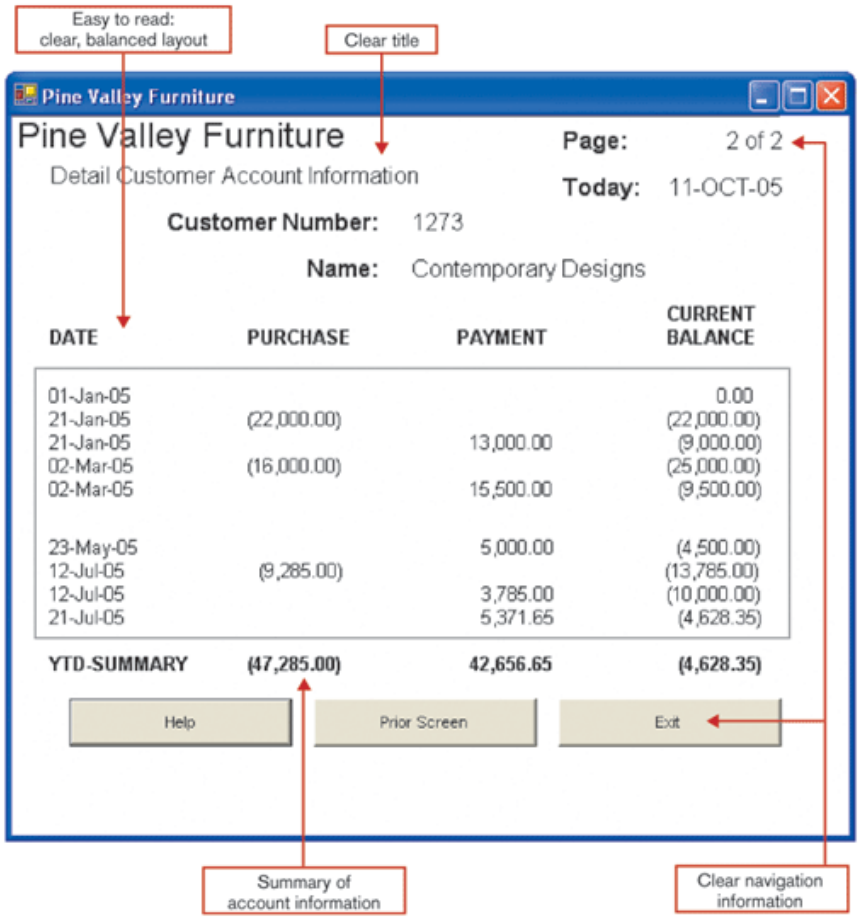
Formatting Forms and Reports (Cont.)

Figure 11-5a Contrasting customer information forms (Pine Valley Furniture) - Poorly designed form



Formatting Forms and Reports (Cont.)

Figure 11-5b Contrasting customer information forms (Pine Valley Furniture) - Improved design for form



Highlighting Information

Notify users of errors in data entry or processing.

Provide warnings regarding possible problems.

Draw attention to keywords, commands, high-priority messages, unusual data values.

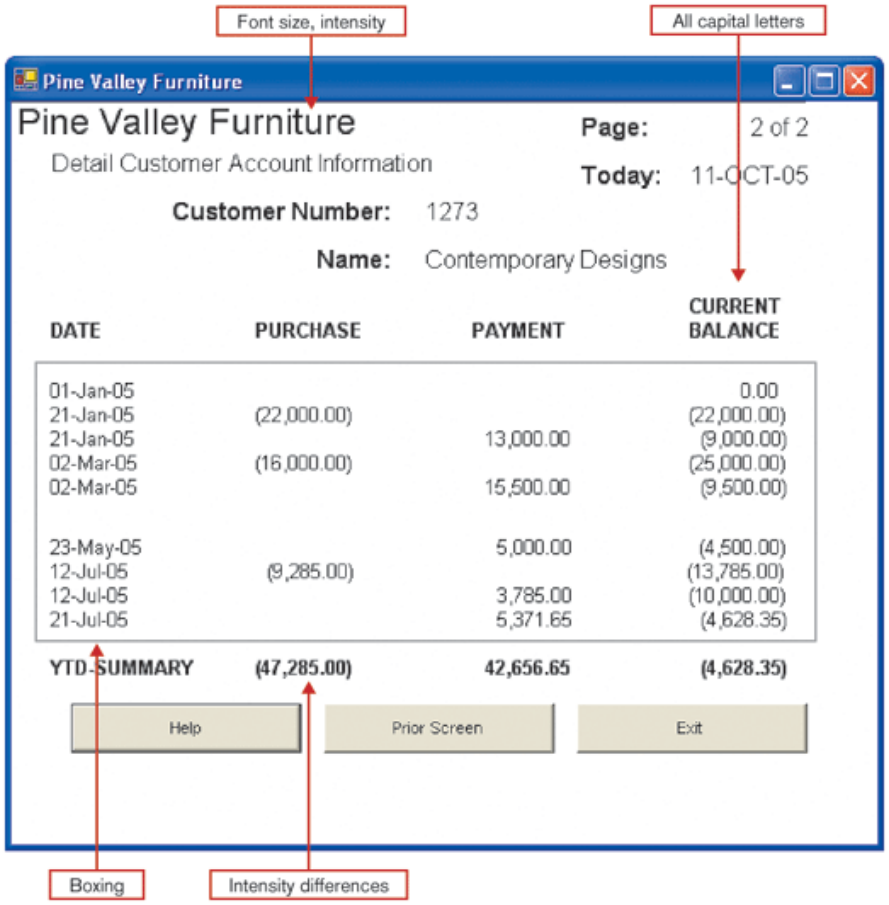
Highlighting Information (Cont.)

Highlighting can include:

- use of upper case, font size differences, bold, italics, underline, boxing, and all capital letters.
- Use of blinking, reverse video, audible tones, and intensity differences.
- And other approaches.

Highlighting Information (Cont.)

Figure 11-6 Customer account status display using various highlighting techniques (Pine Valley Furniture)



Color vs. No Color

Benefits from Using Color

- Soothes or strikes the eye.
- Accents an uninteresting display.
- Facilitates subtle discriminations in complex displays.
- Emphasizes the logical organization of information.
- Draws attention to warnings.
- Evokes more emotional reactions.

Color vs. No Color (Cont.)

Problems from Using Color

- Color pairings may wash out or cause problems for some users.
- Resolution may degrade with different displays.
- Color fidelity may degrade on different displays.
- Printing or conversion to other media may not easily translate.

Displaying Text

Case: mixed upper and lower case, use conventional punctuation.

Spacing: double spacing if possible, otherwise blank lines between paragraphs.

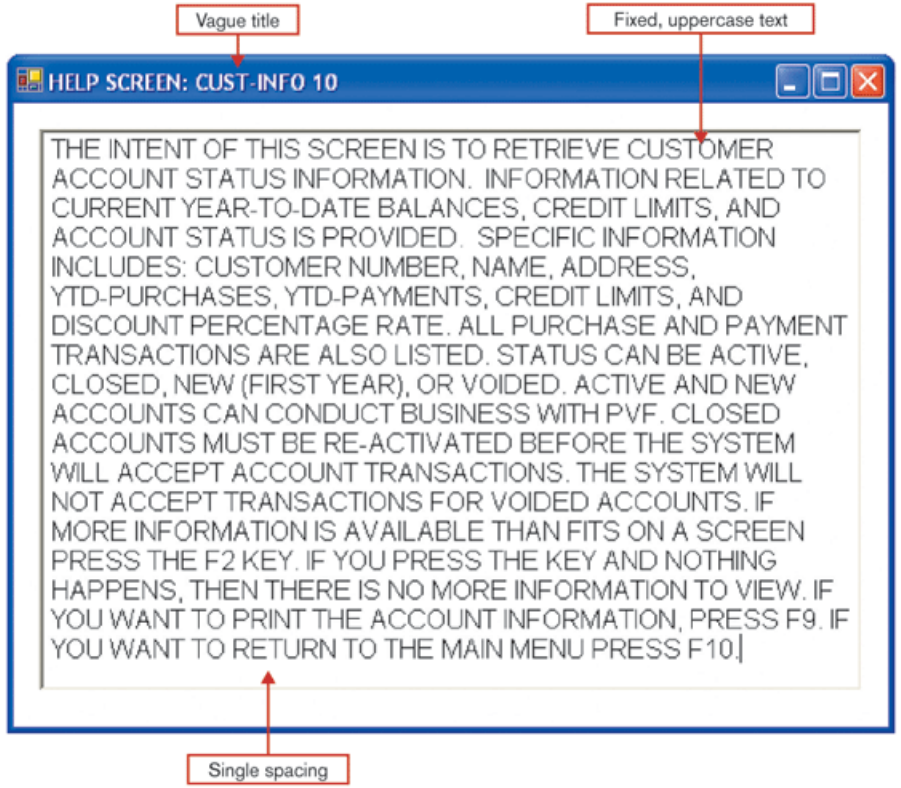
Justification: left justify text, ragged right margins.

Hyphenation: no hyphenated words between lines.

Abbreviations: only when widely understood and significantly shorter than full text.

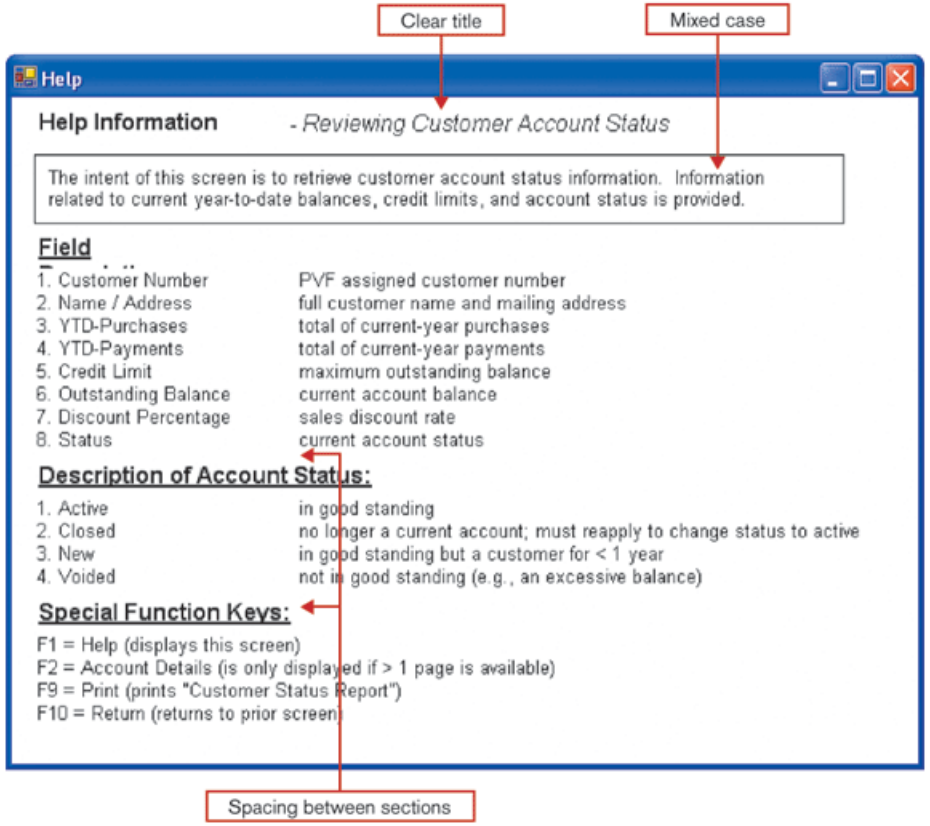
Displaying Text (Cont.)

Figure 11-7a Contrasting the display of textual help information - Poorly designed help screen with many violations of the general guidelines for displaying text



Displaying Text (Cont.)

Figure 11-7b Contrasting the display of textual help information - An improved design for a help screen



Designing Tables and Lists

Labels

- All columns and rows should have meaningful labels.
- Labels should be separated from other information by using highlighting.
- Redisplay labels when the data extend beyond a single screen or page.

Designing Tables and Lists (Cont.)

Formatting columns, rows and text:

- Sort in a meaningful order.
- Place a blank line between every five rows in long columns.
- Similar information displayed in multiple columns should be sorted vertically.

Designing Tables and Lists (Cont.)

- Columns should have at least two spaces between them.
- Allow white space on printed reports for user to write notes.
- Use a single typeface, except for emphasis.
- Use same family of typefaces within and across displays and reports.
- Avoid overly fancy fonts.

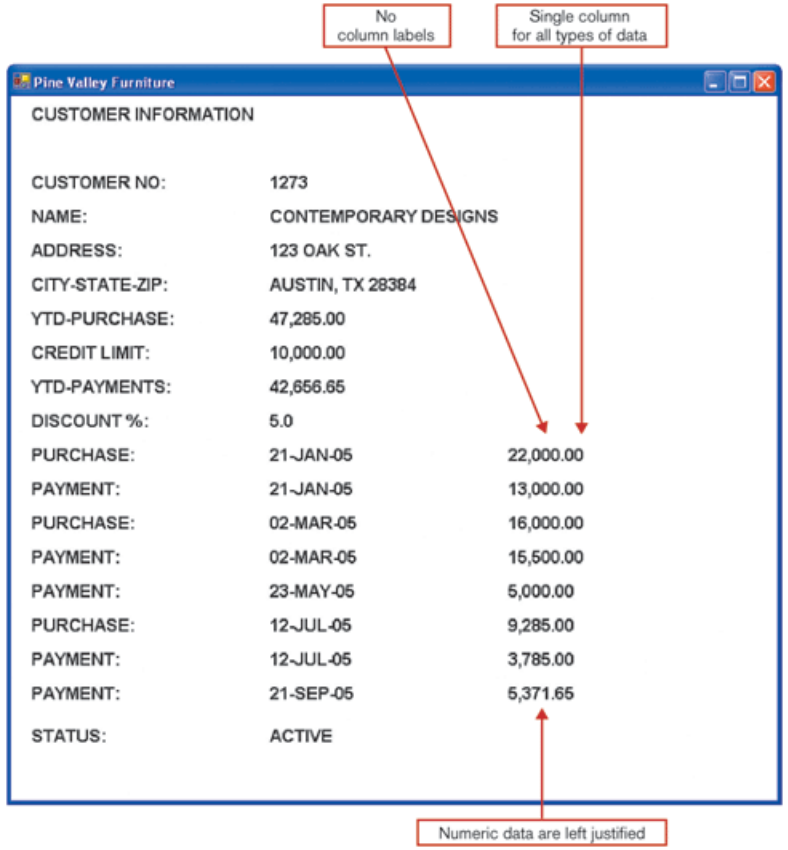
Designing Tables and Lists (Cont.)

Formatting numeric, textual and alphanumeric data:

- Right justify numeric data and align columns by decimal points or other delimiter.
- Left justify textual data. Use short line length, usually 30 to 40 characters per line.
- Break long sequences of alphanumeric data into small groups of three to four characters each.

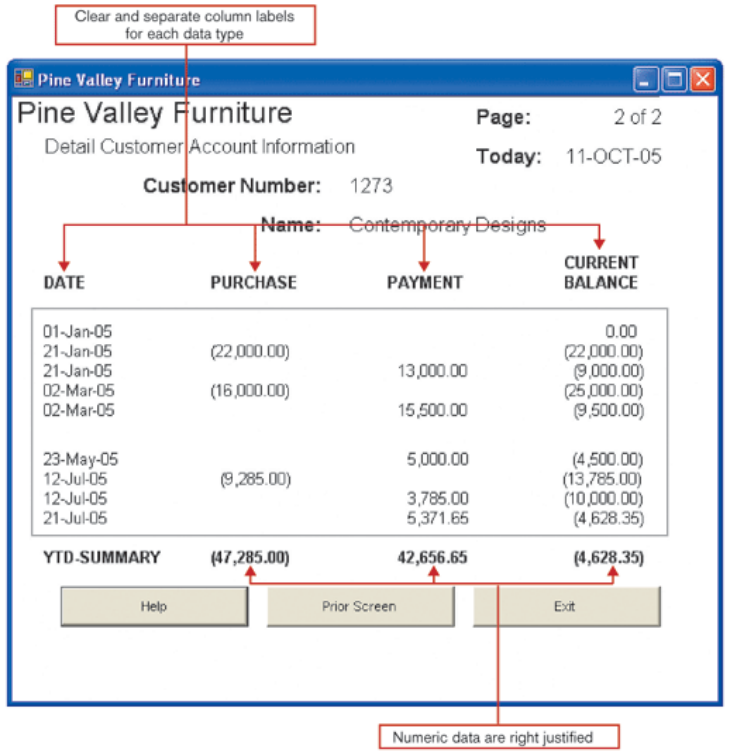
Designing Tables and Lists (Cont.)

Figure 11-8a Contrasting the display of tables and lists (Pine Valley Furniture) - Poorly designed form



Designing Tables and Lists (Cont.)

Figure 11-8b Contrasting the display of tables and lists (Pine Valley Furniture) - Improved design for form



Designing Tables and Lists (Cont.)

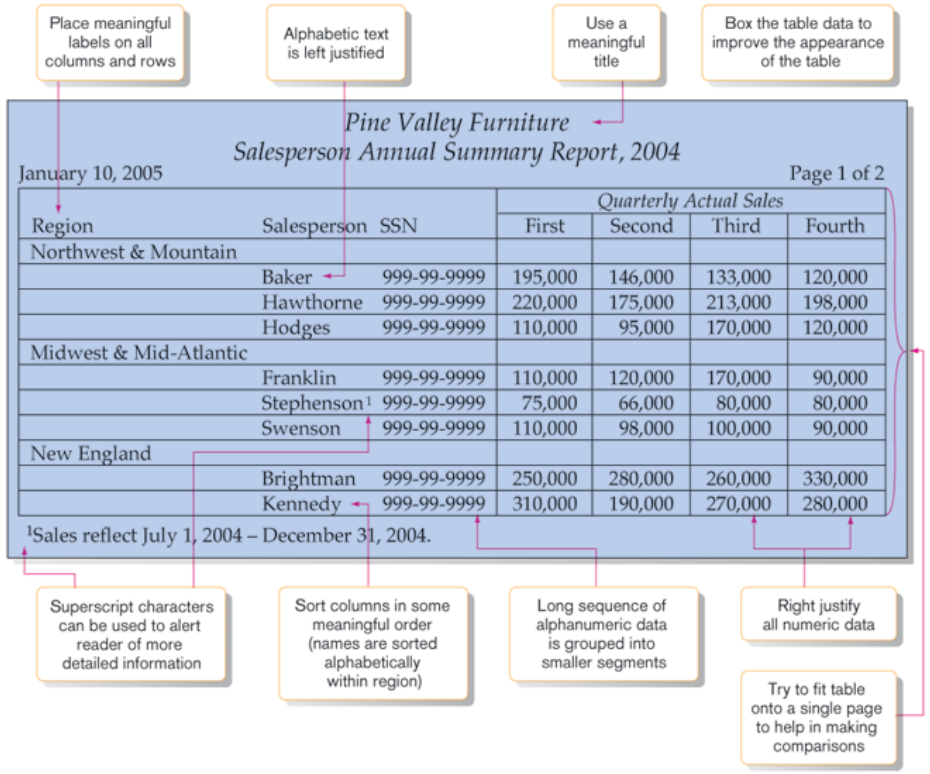
Use tables for reading individual data values.

Use graphs for:

- Providing quick summary.
- Displaying trends over time.
- Comparing points and patterns of variables.
- Forecasting activity.
- Simple reporting of vast quantities of information.

Designing Tables and Lists (Cont.)

Figure 11-9 Tabular report illustrating numerous design guidelines (Pine Valley Furniture)



Designing Tables and Lists (Cont.)

Figure 11-10a Graphs for comparison - Line graph

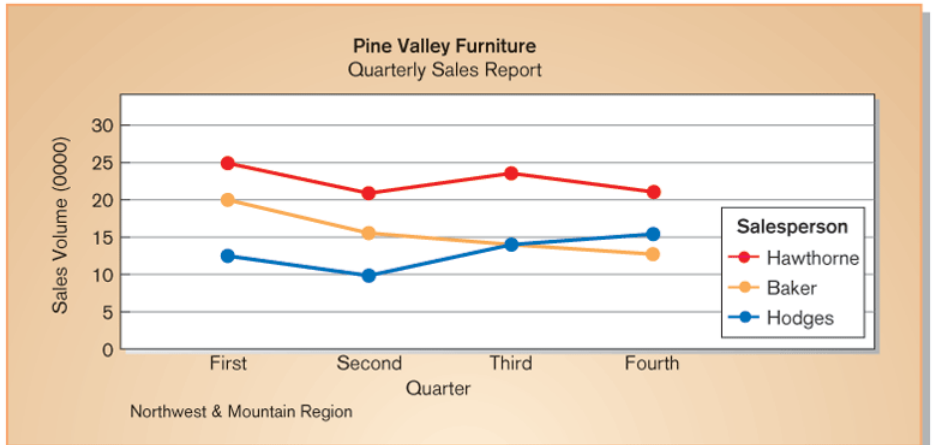
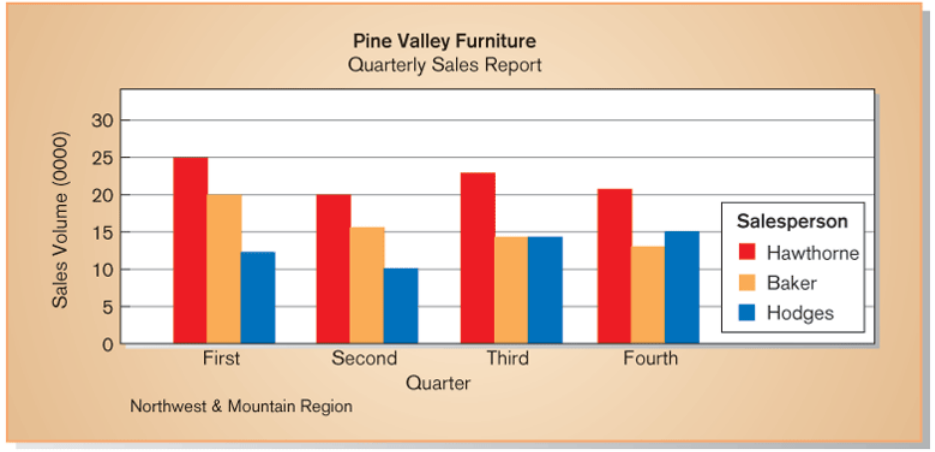


Figure 11-10b Graphs for comparison - Bar graph



Assessing Usability

Objective for designing forms, reports and all human-computer interactions is usability.

There are three characteristics:

- *Speed*. Can you complete a task efficiently?
- *Accuracy*. Does the output provide what you expect?
- *Satisfaction*. Do you like using the output?

Assessing Usability (Cont.)

Usability: an overall evaluation of how a system performs in supporting a particular user for a particular task.

Usability Success Factors

Consistency: of terminology, formatting, titles, navigation, response time.

Efficiency: minimize required user actions.

Ease: self-explanatory outputs and labels.

Format: appropriate display of data and symbols.

Flexibility: maximize user options for data input according to preference.

Usability Success Factors (Cont.)

Characteristics for consideration:

- **User**: experience, skills, motivation, education, personality.
- **Task**: time pressure, cost of errors, work durations.
- **System**: platform.
- **Environment**: social and physical issues.

Measures of Usability

Time to learn.

Speed of performance.

Rate of errors.

Retention over time.

Subjective satisfaction.

Layout of information should be consistent, both within and across applications.

Measures of Usability (Cont.)

Layout of information should be consistent:

- both within and across applications.

Whether information is delivered:

- On screen display or on a hard-copy report.